SIGNIFICANCE TESTS

master recipe • also used in Units 4 and 5

ZONE A · THE FORMULAS + THE FOUR-STEP FRAME

$$z = \frac{\text{statistic} - \text{parameter under } H_0}{\text{standard error of the statistic}}$$

SE estimates the statistic's sampling SD; use the test's null model. A t-procedure uses t with its **df**.

$$P\text{-value} = P(\text{observed or more extreme statistic} \mid H_0 \text{ true})$$

THE SAME FOUR STEPS AS A CONFIDENCE INTERVAL

1	**STATE**	Define the parameter. Give H_0, H_a, and α in context.
2	**PLAN**	Name the test. Check Random → 10% → Normal / Large Counts.
3	**DO**	Statistic, df if needed, and P-value. Draw and shade the curve.
4	**CONCLUDE**	Compare P with α; state evidence for H_a in context.

THE DECISION

$P \le \alpha$
$\longrightarrow$ reject H_0
Convincing evidence for H_a.

$P > \alpha$
$\longrightarrow$ fail to reject H_0
Not convincing evidence for H_a.

ZONE B · ERRORS + POWER

DECISION × TRUE STATE

	H_0 true	H_a true
Reject H_0	Type I error	Correct
Fail to reject H_0	Correct	Type II error

α: significance level.

β and power depend on a specific true effect.

$$\alpha = P(\text{Type I error}) = P(\text{reject } H_0 \mid H_0 \text{ true})$$

$$\beta = P(\text{Type II error}) = P(\text{fail to reject } H_0 \mid H_a \text{ true})$$

Power $= 1 - \beta$

Increases with larger n, larger α, or a larger true effect, with other factors held fixed.

ZONE C · SAY IT IN WORDS

"Because the P-value of ____ is [less/greater] than α = ____, we [reject / fail to reject] H_0. There [is / is not] convincing evidence that ____________________."

Fill the last blank with H_a in context. At $P = \alpha$, use "equal to" and reject.

ZONE D · WATCH OUT

KEEP THE DECISION PRECISE

Say "fail to reject H_0."
Never "accept H_0."

NAME THE PARAMETER

Hypotheses use p or μ,
not $\hat{p}$ or $\bar{x}$.

CHOOSE TAILS BEFORE DATA

One-sided vs two-sided changes P. Let the question choose the tail.

see WS old u6_l4–7, l10–11 · u7_l4–5, l8–10